

BRAC UNIVERSITY
Merul Badda, Dhaka, Bangladesh
CSE331 : Automata and Computability
Assignment 1
Summer 2024

1. Construct DFA for the following regular languages:

$L_1(M) \rightarrow \{w \in \Sigma^* \mid w \text{ doesn't contain } 00\}, \text{ where } \Sigma = \{0, 1\}.$

$L_2(M) \rightarrow \{w \in \Sigma^* \mid w \text{ doesn't contain } 11\}, \text{ where } \Sigma = \{0, 1\}.$

- A. $L(M) \rightarrow (L_1 \cap L_2)'$
- B. $L(M) \rightarrow \{w \in \Sigma^* \mid \text{the sum of the symbols of } w \text{ is a multiple of } 3\}, \text{ where } \Sigma = \{0, 1, 2\}.$
- C. $L(M) \rightarrow \{w \in \Sigma^* \mid \text{the decimal equivalent of } w \text{ is a multiple of } 5\}, \text{ where } \Sigma = \{0, 1\}.$
- D. $L(M) \rightarrow \{w \in \Sigma^* \mid w \text{ is any string not in } 0^*1^*\}, \text{ where } \Sigma = \{0, 1\}.$

2. Write the RE for the following regular languages:

$L_1(M) \rightarrow \{w \in \Sigma^* \mid \text{every third position in } w \text{ is } 1\}, \text{ where } \Sigma = \{0, 1\}.$

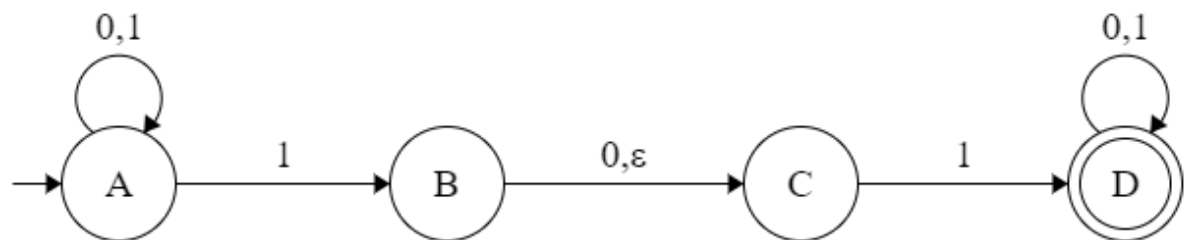
$L_2(M) \rightarrow \{w \in \Sigma^* \mid \text{every } 1 \text{ in } w \text{ is followed by at least two } 0\}, \text{ where } \Sigma = \{0, 1\}.$

- A. $L(M) \rightarrow L_1 \cap L_2$
- B. $L(M) \rightarrow \{w \in \Sigma^* \mid w \text{ starts and ends with the same symbol}\}, \text{ where } \Sigma = \{0, 1\}.$
- C. $L(M) \rightarrow \{w \in \Sigma^* \mid w \text{ contains equal numbers of } 01 \text{ and } 10\}, \text{ where } \Sigma = \{0, 1\}.$
- D. $L(M) \rightarrow \{w \in \Sigma^* \mid w \text{ does not contain } 101\}, \text{ where } \Sigma = \{0, 1\}.$

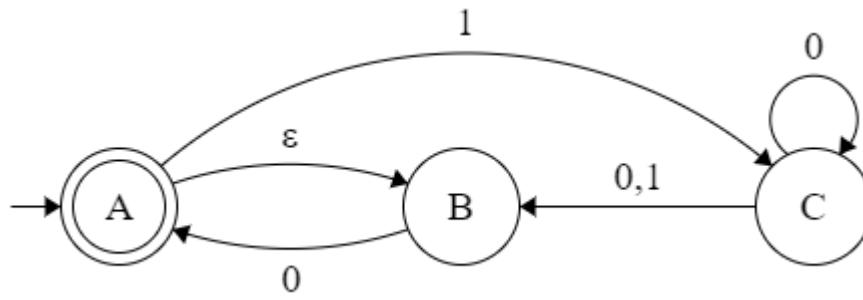
3. Construct NFA for the following regular languages:

- A. $L(M) \rightarrow \{w \in \Sigma^* \mid w \text{ contains } 1001 \text{ or } 11\}, \text{ where } \Sigma = \{0, 1\}. \text{ (use 5 states.)}$
- B. $L(M) \rightarrow \{w \in \Sigma^* \mid w \text{ contains a } 1 \text{ in the third position from the end}\}, \text{ where } \Sigma = \{0, 1\}.$
- C. $L(M) \rightarrow \{w \in \Sigma^* \mid \text{length of } w \text{ is a multiple of } 2 \text{ or } 3\}, \text{ where } \Sigma = \{0, 1\}.$

4. Convert the following NFA into DFA:



A.



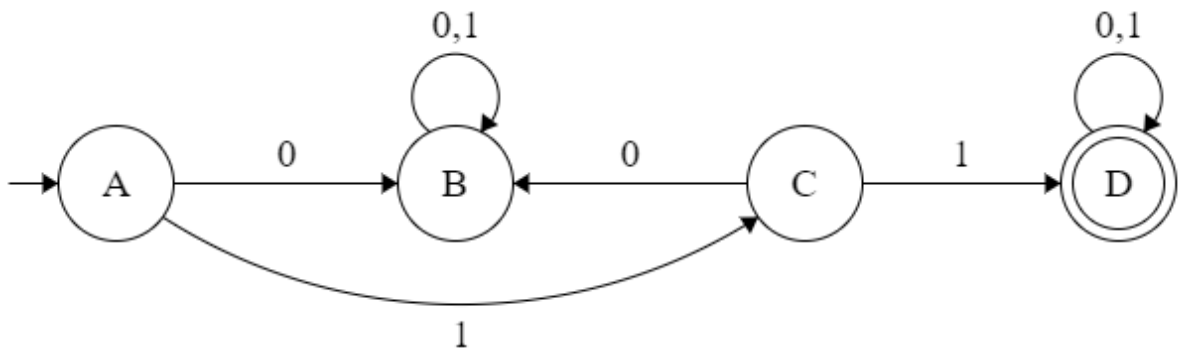
B.

5. Convert the following RE into NFA:

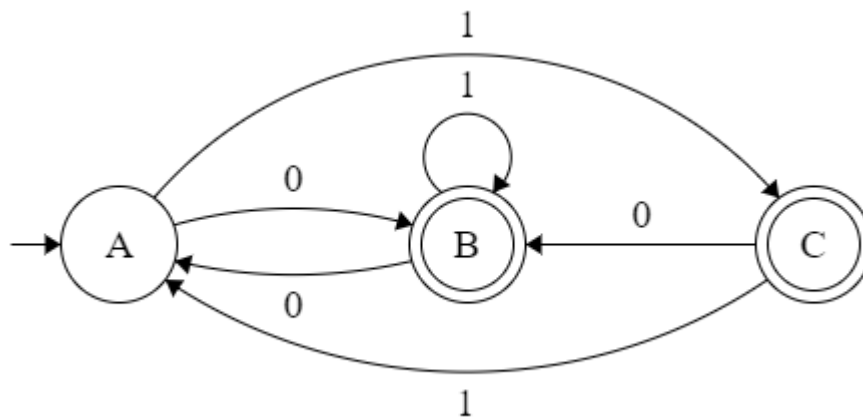
A. $(01 \mid 0)^*$

B. $(0 \mid 1)^*010$

6. Convert the following DFA into RE:



A.



B.

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Assignment 02

Summer 2024

Q1. Which of the following statements about regular expressions is true? (2 points)

1. Regular expressions can describe only Regular Languages.
2. Regular expressions can describe only Context Free Languages.
3. Regular expressions can describe both Regular Language and Context Free Languages.

Write an explanation for your answer.

Q2. Which is the correct regular expression for the language $L = \{w \in \{0,1\}^*: 0^n, \text{ where } n \text{ is even}\}$? (2 points)

1. $0(00)^*$
2. $(00)^*$
3. $0^*(00)^*0^*$
4. None of the above

Write an explanation for your answer.

Q3. Which is the correct regular expression for the language $L = \{w \in \{0,1\}^*: 0^n1^n, \text{ where } n \text{ is even}\}$? (2 points)

1. $(00)^*(11)^*$
2. $((0 \cup 1)(0 \cup 1))^*$
3. $(00 \cup 11)^*$
4. None of the above

Write an explanation for your answer.

Q4. You are given two languages, A and B. You can construct a regular expression for A. However, it is impossible to construct a regular expression for B. Now read the options below. (2 points)

- a) A is a regular language
- b) B is a regular language.
- c) B is a nonregular language.
- d) $A \cup B$ is a regular language. Choose the best answer from below.
 - 1. a and b
 - 2. a and c
 - 3. a and d
 - 4. a, b and d
 - 5. a, c and d

Write an explanation for your answer.

Q5. What does the regular expression $(0 \cup 1)^* 0 (0 \cup 1)^*$ describe? (2 points)

- 1. $L = \{w \in \{0,1\}^*: w \text{ contains exactly one } 0\}$
- 2. $L = \{w \in \{0,1\}^*: w \text{ contains at least three } 0\}$
- 3. $L = \{w \in \{0,1\}^*: w \text{ contains at least one } 0\}$
- 4. $L = \{w \in \{0,1\}^*: w \text{ contains at most one } 0\}$ Write an explanation for your

answer.

Q6. $(0 \cup 1)^* (0 (0 \cup 1) 0 \cup 1 (100 \cup 001 \cup 11^*)) 00$

Write down the shortest nonempty string generated by the given expression. Write down the string only. (2 points)

Write an explanation for your answer.

Q7. $(00 \cup 01) (101)^* ((00)^* \cup 110)$

What is the shortest string generated by the regular expression with exactly three 0s? Write down the string. (2 points)

Write an explanation for your answer.

Q8. Does the string 00100001011 belong to the regular expression $0(0 \cup 1)^*01(0 \cup 1)^*((0 \cup 1)^*)^*11(0 \cup 1)^*1$? (2 points)

1. Yes
2. No

Write an explanation for your answer.

Q9. Let's, $\Sigma = \{a, b\}$. You are given a regular expression, $a(\Sigma^*a^+\epsilon) \cup b(\Sigma^*b^+\epsilon)$, what is the correct language for this regex? Here, $a^+ = aa^*$.

Note, for a regular language L , the regular expression will be correct if and only if it generates all the strings, $w \in L$, and doesn't generate any string, $w \notin L$. (2 points) a) w starts and ends with the same symbols

b) w contains either at least two a or at least two b

c) w contains substring ab

d) w contains an a followed by a b

e) w contains equal numbers of ab and ba Choose the best answer from below.

1. a
2. a and b
3. c and d
4. a and e
5. a, b and e

Write an explanation for your answer.

Q10. Which of the following Regular expressions are equivalent to $a(a^*b^* \cup b^+)b$ (2 points) a)

$a^*(ab^+)$

b) $a^+(ab^+)$

c) a^+b^+

d) $a(aa^*b^*)b$

e) $a^*(a^*b^+ \cup b^*)b$

Choose the best answer from below.

1. a and b

2. c and d
3. d and e
4. b and c

Write an explanation for your answer.

Q11. $(01 \cup 1010 \cup 1)^* (00 \cup 010)$

The length of a string, w can be expressed by $|w|$. Find out how many distinct strings can be generated using the given regular expression such that $3 \leq |w| \leq 5$. Write a numeric value only. (2 points)

Write an explanation for your answer.

Q12. You have a regular expression $(0 \cup 1)(0 \cup 1)00^*(0 \cup 1)^*1^*$. And another regular expression $(00 \cup 11 \cup 10 \cup 01)01^*0^*(1 \cup 0 \cup \epsilon)^*0^*$. Are these two the same? (2 points)

1. Yes
2. No

Write an explanation for your answer.

Q13. Let's $\Sigma = \{0,1\}$. What is the regular expression for the strings that have even length and start with 1? (2 points) a) $1(0 \cup 1)^*$

b) $(10 \cup 11)((0 \cup 1)(0 \cup 1))^*$

c) $(1(0 \cup 1))^*$

d) $1(0 \cup 1)((0 \cup 1)(0 \cup 1))^*$

e) $1(0 \cup 1)(00 \cup 01 \cup 11 \cup 10)^* \cup \epsilon$ Select the best answer.

1. d and e
2. b and e
3. b and d
4. b, d and e
5. a, c and e

Write an explanation for your answer.

Q14. $L = \{w \in \{0,1\}^*: w \text{ doesn't contain } 001\}$

Note, for a regular language L , the regular expression will be correct if and only if it generates all the strings, $w \in L$, and doesn't generate any string, $w \notin L$. Which of the following Regular Expressions generate L :

- $(1^*(01)^*)^*(\epsilon \cup 0 \cup 000^*)$ (1 point)
 1. Correct
 2. Incorrect
- $(1^*(0 \cup \epsilon)1^*)^*$ (1 point)
 1. Correct
 2. Incorrect
- $(1 \cup 10)^*(\epsilon \cup 0 \cup 000^*)$ (1 point)
 1. Correct
 2. Incorrect
- $(1 \cup 01)^*0^*$ (1 point)
 1. Correct
 2. Incorrect
- $(1^*(10)^*)^*0^*$ (1 point)
 1. Correct
 2. Incorrect

Q15. $(0 \cup \epsilon)(10)^*(1 \cup \epsilon)$

Note, for a regular language L , the regular expression will be correct if and only if it generates all the strings, $w \in L$, and doesn't generate any string, $w \notin L$.

Which of the following regular languages are generated by the given regular expression

- $L = \{w \in \{0,1\}^*: w \text{ doesn't contain } 00\}$ (1 point)
 1. Correct
 2. Incorrect
- $L = \{w \in \{0,1\}^*: 0 \text{ and } 1 \text{ alternates in } w\}$ (1 point)
 1. Correct
 2. Incorrect
- $L = \{w \in \{0,1\}^*: w \text{ doesn't contain either } 00 \text{ or } 11\}$ (1 point)
 1. Correct

2. Incorrect

- $L = \{ w \in \{0,1\}^* : \text{The period of } w \text{ is either } 01 \text{ or } 10 \}$

The period of a string is the smallest prefix, when repeated one or more times, can recreate the entire string. (1 point)

1. Correct

2. Incorrect

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Assignment 03

Summer 2024

Q1. Which of the following statements about Context Free Grammar (CFG) is true? (2 points)

1. Context Free Grammar (CFG) can describe only Context Free Languages.
2. Context Free Grammar (CFG) can describe all the Non Regular Languages and Regular Languages.
3. Context Free Grammar (CFG) can describe both Context Free Languages and Regular Language.

Write an explanation for your answer.

Q2. Consider the following languages.

A) $L = \{ w \in \{a,b,c\}^* : a^n b^n c^n, \text{ where } n \geq 0 \}$

B) $L = \{ w \in \{0,1\}^* : ww \}$

C) $L = \{ w \in \{0,1\}^* : ww^R \}$ [w^R means reverse of w . For example, if $w = 10110$, then $w^R = 01101$.]

D) $L = \{ w \in \{a,b,c\}^* : a^i b^j c^k, \text{ where } 0 \leq i \leq j \leq k \}$

Which languages are Context Free Languages?

1. Only C
2. B and C
3. C and D
4. A, B and C

Write an explanation for your answer.

Q3. Let, L is a context-free language. Then L^* will also be a context-free language.

The statement is

1. True
2. False
3. Not enough information to answer

Write an explanation for your answer.

Q4. Consider the following Context Free Grammar.

$S \rightarrow AB \mid BC$

$A \rightarrow aA \mid Aa \mid a \mid \varepsilon$

$B \rightarrow bBb \mid bb$

$C \rightarrow c \mid \varepsilon$

Choose the correct regular expression for the language covered by the given CFG

1. $a^*(bb)^* \mid (bb)^*c^*$
2. $a^+(bb)^* \mid (bb)^+c^?$
3. $a^+(bb)^+ \mid (bb)^+(c|\varepsilon)$
4. $a^*(bb)^+ \mid (bb)^+(c|\varepsilon)$
5. $a^+(bb)^* \mid (bb)^*(c|\varepsilon)$

Write an explanation for your answer.

Q5. $A \rightarrow 1A \mid 0A1 \mid 01$

Which of the following string can be generated by the given CFG.

There could be more than one correct answer, choose all of them.

1. 000111
2. 001111
3. 001011
4. 0100111
5. 1000111

Write an explanation for your answer.

Q6. You are given a Context Free Grammar.

$S \rightarrow AB$

$A \rightarrow aA \mid \varepsilon$

$B \rightarrow Bb \mid \varepsilon$

Now consider the following languages.

- a) $L = \{ w \in \{a,b\}^*: a^n b^m, \text{ where } n, m \geq 0 \}$
- b) $L = \{ w \in \{a,b\}^*: a^n b^n, \text{ where } n \geq 0 \}$
- c) $L = \{ w \in \{a,b\}^*: \text{number of } a \text{ and number of } b \text{ is unequal in } w \}$
- d) $L = \{ w \in \{a,b\}^*: \text{all } a \text{ in } w \text{ always precede } b \}$

Which of the following statements are true regarding the language generated by this grammar?

Note, for a language L , the CFG will be correct if and only if it can parse all the strings, $w \in L$, and doesn't parse any string, $w \notin L$.

1. Only A
2. Only B
3. A and C
4. A and D
5. B and D

Write an explanation for your answer.

Q7. $L = \{ w \in \{a,b,c\}^*: a^m b^n c^k \mid n, m \geq 0 \text{ \& } k=3m+2 \}$

Let's say, we have the following four CFGs labeled as (A) to (D).

CFG A:

$$S \rightarrow aScc \mid Y$$
$$Y \rightarrow bY \mid ccc$$

CFG B:

$$S \rightarrow aSccc \mid Y$$
$$Y \rightarrow bY \mid cc$$

CFG C:

$$S \rightarrow Xcc$$
$$X \rightarrow aXccc \mid Z$$
$$Z \rightarrow bZ \mid \varepsilon$$

CFG D:

$$S \rightarrow aSccc \mid bS \mid cc$$

What will be the correct CFG for the language L ?

Note, for a language L , the CFG will be correct if and only if it can parse all the strings, $w \in L$, and doesn't parse any string, $w \notin L$.

1. Only B
2. Only C
3. A and C
4. A and D

5. B and C
6. B and D

Write an explanation for your answer.

Q8. $L = \{ w \in \{a,b\}^* : w = a^i b^j \text{ where } i = 2+j, j \geq 0 \}$

Let's say, we have the following four CFGs labeled as (A) to (C).

CFG A:

$S \rightarrow aaN$

$N \rightarrow aNb \mid \epsilon$

CFG B:

$S \rightarrow aSb \mid aa$

CFG C:

$S \rightarrow aSb \mid N$

$N \rightarrow aaN \mid \epsilon$

What will be the correct CFG for the language L?

Note, for a language L, the CFG will be correct if and only if it can parse all the strings, $w \in L$, and doesn't parse any string, $w \notin L$.

1. A and C
2. A and B
3. B and C
4. A, B and C

Write an explanation for your answer.

Q9. Consider the following two Context Free Grammars. Do these two grammars (G1 and G2) represent the same language?

G1:

$S \rightarrow AB$

$A \rightarrow 1A00 \mid C$

$C \rightarrow 0C \mid 0$

$B \rightarrow 1B \mid \epsilon$

G2:

$S \rightarrow ABC$

$A \rightarrow 1A00 \mid \epsilon$

$B \rightarrow 0B \mid 0$

$C \rightarrow 1C \mid \epsilon$

1. Yes
2. No

Write an explanation for your answer.

Q10. $L = \{ w \in \{a,b,c\}^* : a^m b^n c^k \mid n = m/2 \text{ or } n = k - 2 ; \text{ where } n \geq 0, m \text{ is even \& } k \geq 2 \}$

Let's say, we have the following 4 CFGs labeled as (A) to (D).

CFG A:

$S \rightarrow XY \mid PQ$

$X \rightarrow aXbb \mid \varepsilon$

$Y \rightarrow ccY \mid c$

$P \rightarrow aP \mid b$

$Q \rightarrow bQc \mid c$

CFG B:

$S \rightarrow XY \mid PQ$

$X \rightarrow aXb \mid \varepsilon$

$Y \rightarrow cYb \mid c$

$P \rightarrow aP \mid \varepsilon$

$Q \rightarrow bQcc \mid c$

CFG C:

$S \rightarrow XY \mid PQ$

$X \rightarrow aXb \mid c$

$Y \rightarrow cYb \mid \varepsilon$

$P \rightarrow aP \mid \varepsilon$

$Q \rightarrow bQcc \mid c$

CFG D:

$S \rightarrow XY \mid PQ$

$X \rightarrow aaXb \mid \varepsilon$

$Y \rightarrow cY \mid cc$

$P \rightarrow aaP \mid \varepsilon$

$Q \rightarrow bQc \mid cc$

What will be the correct CFG for the language L?

Note, for a language L , the CFG will be correct if and only if it can parse all the strings, $w \in L$, and doesn't parse any string, $w \notin L$.

1. A
2. B
3. C
4. D

Write an explanation for your answer.

Q11. $L = \{w \in \{0,1\}^*: w_1w_2w_3: \text{ where } |w_1| = |w_2| \text{ or } |w_3| = |w_1|\}$

Let's say, we have the following 4 CFGs labeled as (A) to (D).

CFG A:

$S \rightarrow RQ \mid P$

$P \rightarrow 0P1 \mid 1P0 \mid \varepsilon$

$Q \rightarrow 0Q \mid 1Q \mid \varepsilon$

$R \rightarrow 0R0 \mid 1R1 \mid 0R1 \mid 1R0 \mid Q$

CFG B:

$S \rightarrow PR \mid Q$

$P \rightarrow 0P0 \mid 1P1 \mid \varepsilon$

$Q \rightarrow 0Q \mid 1Q \mid \varepsilon$

$R \rightarrow 0R0 \mid 1R1 \mid 0R1 \mid 1R0 \mid Q$

CFG C:

$S \rightarrow PQ \mid R$

$P \rightarrow 0P0 \mid 1P1 \mid 0P1 \mid 1P0 \mid \varepsilon$

$Q \rightarrow 0Q \mid 1Q \mid \varepsilon$

$R \rightarrow 0R0 \mid 1R1 \mid 0R1 \mid 1R0 \mid Q$

CFG D:

$S \rightarrow PQ \mid R$

$P \rightarrow 0P0 \mid 1P1 \mid 0P1 \mid 1P0 \mid \varepsilon$

$Q \rightarrow 0Q \mid 1Q$

$R \rightarrow 0R0 \mid 1R1 \mid 0R1 \mid 1R0 \mid P$

What will be the correct CFG for the language L ?

Note, for a language L , the CFG will be correct if and only if it can parse all the strings, $w \in L$, and doesn't parse any string, $w \notin L$.

1. A and C

2. B and C
3. C and D
4. A, B and C
5. A, B, C and D

Write an explanation for your answer.

Q12. $L = \{w \in \{0,1\}^* : w_1\#w_2\#w_3 : \text{where } w_3 = w_1^R\}$

Given a string w over some alphabet Σ , let w^R be its reverse. For example, if $w = 10110$, then $w^R = 01101$.

Let's say, we have the following 4 CFGs labeled as (A) to (C).

CFG A:

$S \rightarrow 1S0 \mid 0S1 \mid \#P\#$

$P \rightarrow 0P \mid 1P \mid \epsilon$

CFG B:

$S \rightarrow 0S \mid 1S \mid \#P\#$

$P \rightarrow 0P0 \mid 1P1 \mid \epsilon$

CFG C:

$S \rightarrow 0S0 \mid 1S1 \mid \#P\#$

$P \rightarrow 0P \mid 1P \mid \epsilon$

What will be the correct CFG for the language L ?

Note, for a language L , the CFG will be correct if and only if it can parse all the strings, $w \in L$, and doesn't parse any string, $w \notin L$.

1. A
2. B
3. C

Write an explanation for your answer.

Q13. $L = \{w \in \{0,1\}^* : w \in \Sigma^* \mid \text{every prefix of } w \text{ has at least as many 0's as 1's}\}$

Clarification: L states "Every prefix of w contains equal or more 0s than 1s."

Select the correct CFG for L :

A) $S \rightarrow 0S1S \mid 0S \mid \epsilon$

$$B) S \rightarrow SS \mid 0S1 \mid 1 \mid \varepsilon$$

$$C) S \rightarrow SS \mid 0S1 \mid 0 \mid \varepsilon$$

$$D) S \rightarrow 0S1S \mid 1S \mid \varepsilon$$

A prefix is a string consisting of several first letters of a given string, without any reorders. An empty prefix is also a valid prefix. For example, the string "abcd" has 5 prefixes: empty string, "a", "ab", "abc" and "abcd".

1. a and d
2. a and b
3. a and c
4. a, b and d
5. a, b and c
6. a, b, c and d

Write an explanation for your answer.

Q14. Let $\Sigma = \{a, b\}$ and let $L = \{a^n b^m \mid n, m \in \mathbb{N} \text{ and } n \leq m \leq 5n\}$. The CFG for L is given below.

Here is a CFG of the language

$$S \rightarrow aSb? \mid \varepsilon$$

$$A \rightarrow b \mid \varepsilon$$

What will be the '?' such that the grammar can describe the language, L correctly.

Fill out the question mark with one or more variables/terminals. Write down the missing string only.

Write an explanation for your answer.

Q15.

$$E \rightarrow E + P \mid P$$

$$P \rightarrow P * Q \mid Q$$

$$Q \rightarrow \text{id}$$

Is the grammar ambiguous?

1. Yes
2. No

Write an explanation for your answer.

Q16. You are given two Context Free Grammar G1 and G2.

G1:

$S \rightarrow (S)S \mid \epsilon$

G2:

$S \rightarrow (S) \mid SS \mid \epsilon$

Now, consider the string $w = ((()()))((()()))()$

Which of the following statements is true?

1. w can be generated using only G1
2. w can be generated using only G2
3. w can be generated using G1 and G2
4. w cannot be generated using either G1 or G2

Write an explanation for your answer.

Q17. Consider the two Context free grammars G1 and G2. Which of the following is $L(G1) \cap L(G2)$?

G1:

$S \rightarrow ASA$

$S \rightarrow 0$

$A \rightarrow 0 \mid 1$

G2:

$S \rightarrow ASA$

$S \rightarrow \epsilon$

$A \rightarrow 0 \mid 1$

1. $L = \{w \in \Sigma^* \mid w \text{ contains } 0 \text{ in the middle.}\}$
2. $L = \{w \in \Sigma^* \mid \text{length of } w \text{ is even.}\}$
3. $L = \{w \in \Sigma^* \mid w \text{ is a even length string with } 0 \text{ in the middle.}\}$
4. $L = \{w \in \Sigma^* \mid w \text{ is a odd length string with } 0 \text{ in the middle.}\}$
5. None of the above.

Write an explanation for your answer.

Q18. Consider the following Context Free Grammar, G.

G:

$P \rightarrow XP1 \mid XQ0$

$Q \rightarrow XQ1 \mid XR0$
 $R \rightarrow XRX \mid \#$
 $X \rightarrow 0 \mid 1$

Now, answer the following questions.

What is the shortest string can be parsed from the grammar, G? If there are multiple correct answer write only one of those.

Write an explanation for your answer.

Find out how many distinct strings can be generated using the given Context Free Grammar, G, such that $9 \leq |w| \leq 11$. Write the numeric value only.

Write an explanation for your answer.

Q19. Consider the following Context Free Grammar, G.

G:

$S \rightarrow aaS \mid abS \mid baS \mid bbS \mid X$
 $X \rightarrow aaY \mid baY$
 $Y \rightarrow aY \mid bY \mid \epsilon$

Write a eight length string that starts with "bb" and has only one parse tree. Write the string only.

Write an explanation for your answer.

Q20. $L = \{w \in \{a,b,c\}^* : a^i b^j c^k \mid i, j, k \geq 0 \text{ and if } i > j \text{ then } k = i - j, \text{ else } k = 0\}$

Which of the following CFGs generates L?

CFG A:

$S \rightarrow AB \mid AC$
 $A \rightarrow aA \mid \epsilon$
 $B \rightarrow bB \mid b$
 $C \rightarrow aCc \mid \epsilon$

CFG B:

$S \rightarrow AB \mid A$
 $A \rightarrow aA \mid a \mid \epsilon$
 $B \rightarrow bBc \mid b$

CFG C:

$S \rightarrow A \mid B$
 $A \rightarrow aAb \mid Ab \mid \varepsilon$
 $B \rightarrow aBc \mid aCc$
 $C \rightarrow aCb \mid \varepsilon$

CFG D:

$S \rightarrow AB \mid AC$
 $A \rightarrow aAb \mid \varepsilon$
 $B \rightarrow bB \mid \varepsilon$
 $C \rightarrow aCc \mid \varepsilon$

Note, for a language L , the CFG will be correct if and only if it can parse all the strings, $w \in L$, and doesn't parse any string, $w \notin L$.

1. A
2. B
3. C
4. D

Write an explanation for your answer.

Q21. $L = \{w \in \{a, b\}^* : \text{each } b \text{ in } w \text{ is followed by at least two } a\}$

Note, for a language L , the CFG will be correct if and only if it can parse all the strings, $w \in L$, and doesn't parse any string, $w \notin L$.

Which of the following Context Free Grammar can generate L ?

$S \rightarrow XY$
 $X \rightarrow aX \mid \varepsilon$
 $Y \rightarrow ZY \mid \varepsilon$
 $Z \rightarrow PQ$
 $P \rightarrow ba$
 $Q \rightarrow aQ \mid a$

1. Correct
2. Incorrect

$P \rightarrow aP \mid bQ \mid \varepsilon$
 $Q \rightarrow aR \mid bT$
 $R \rightarrow aS \mid bT$
 $S \rightarrow aS \mid bQ \mid \varepsilon$
 $T \rightarrow aT \mid bT$

1. Correct

2. Incorrect

$S \rightarrow AbR \mid A$

$A \rightarrow aA \mid \varepsilon$

$R \rightarrow aAabR \mid A$

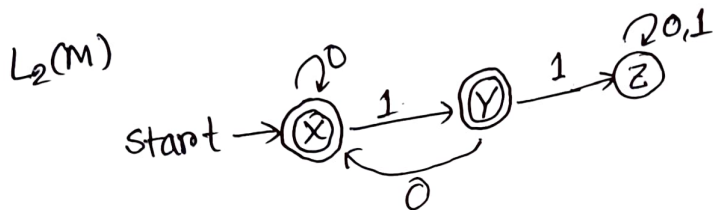
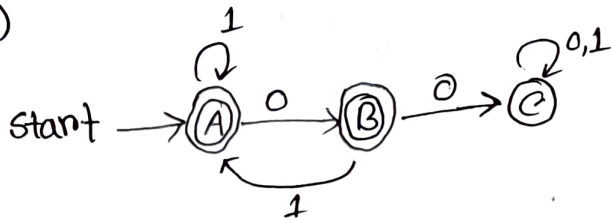
1. Correct

2. Incorrect

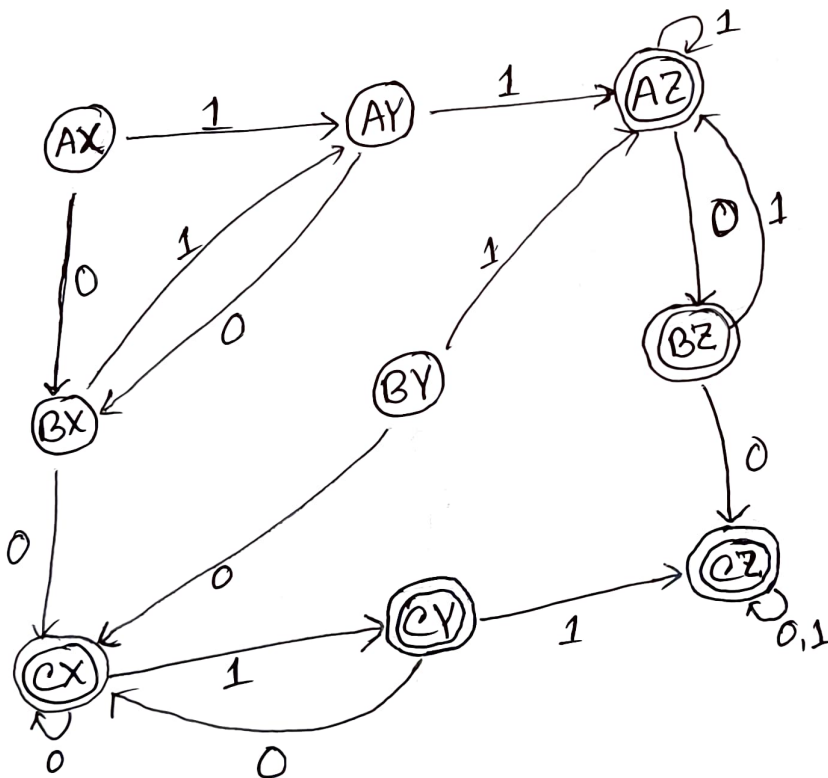
CSE331
ASSIGNMENT 01
SECTION 16
SUMMER'24

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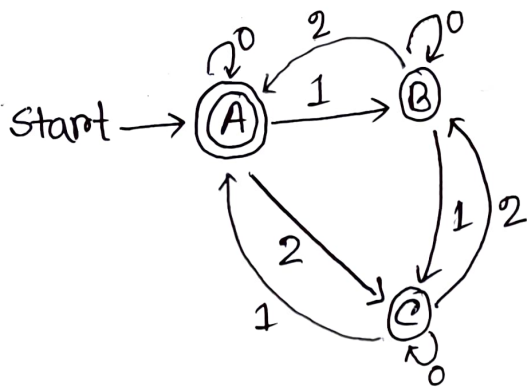
① A $L_1(M)$



$(L_1(M) \cap L_2(M))'$

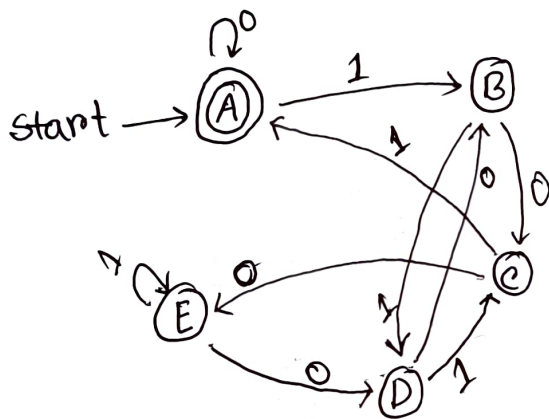


B



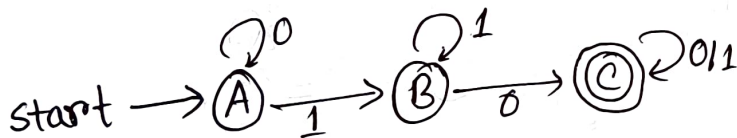
$$\begin{aligned} A &\rightarrow \%3 = 0 \\ B &\rightarrow \%3 = 1 \\ C &\rightarrow \%3 = 2 \end{aligned}$$

C



$$\begin{aligned} A &\rightarrow \%5 = 0 \\ B &\rightarrow \%5 = 1 \\ C &\rightarrow \%5 = 2 \\ D &\rightarrow \%5 = 3 \\ E &\rightarrow \%5 = 4 \\ &(2n+0, 2n+1) \end{aligned}$$

D



②

A $L_1(M) \rightarrow (\Sigma \Sigma 1)^* (1101\epsilon) (1101\epsilon)$

$L_2(M) \rightarrow 0^* (1000^*)^*$

$L(M) \rightarrow L_1 \cap L_2$

$\rightarrow (001)^* 00$

B

$0\Sigma^*0 \mid 1\Sigma^*1 \mid 011$

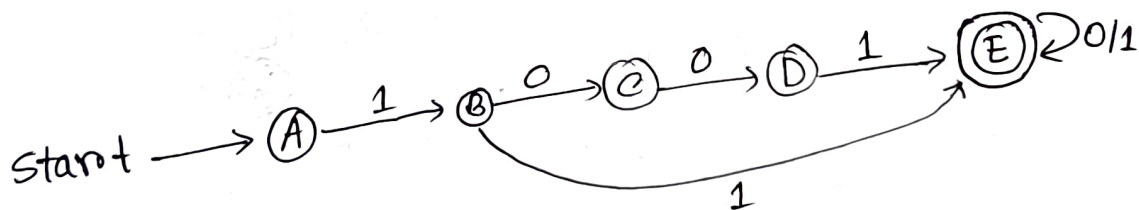
C

$0\Sigma^*0 \mid 1\Sigma^*1 \mid 011$

D $(011)^* (\epsilon \mid 010011 \mid 110)^*$

③

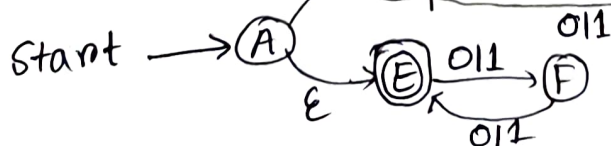
A



B



C



④

A

start state = $\{A\} = S$

$\delta(S, 0) = \{A\} = S$

$\delta(S, 1) = \{A, B\} \xrightarrow{\epsilon} \{A, B, C\} = M$

$\delta(M, 0) = \{A, C\} = N$

$\delta(M, 1) = \{A, B, D\} \xrightarrow{\epsilon} \{A, B, C, D\} = P$

$\delta(N, 0) = \{A\} = S$

$\delta(N, 1) = \{A, B, D\} \xrightarrow{\epsilon} \{A, B, C, D\} = P$

$\delta(P, 0) = \{A, C, D\} = Q$

$\delta(P, 1) = \{A, B, D\} \xrightarrow{\epsilon} \{A, B, C, D\} = P$

$\delta(Q, 0) = \{A, D\} = R$

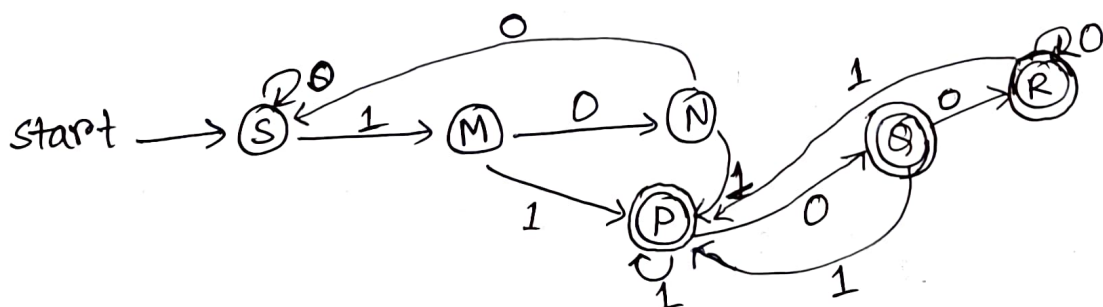
$\delta(Q, 1) = \{A, B, D\} \xrightarrow{\epsilon} \{A, B, C, D\} = P$

$\delta(R, 0) = \{A, D\} = R$

$\delta(R, 1) = \{A, B, D\} \xrightarrow{\epsilon} \{A, B, C, D\} = P$

Total states: $\{A\} = S$; $\{A, B, C\} = M$; $\{A, C\} = N$; $\{A, B, C, D\} = P$

$\{A, C, D\} = Q$; $\{A, D\} = R$



B

Start state = $\{A, B\} = S$

$\delta(S, 0) = \{A\} \xrightarrow{\epsilon} \{A, B\} = S$

$\delta(S, 1) = \{C\} = M$

$\delta(M, 0) = \{B, C\} = N$

$\delta(M, 1) = \{B\} = P$

$\delta(N, 0) = \{A, B, C\} \xrightarrow{\epsilon} \{A, B, C\} = Q$

$\delta(N, 1) = \{B\} = P$

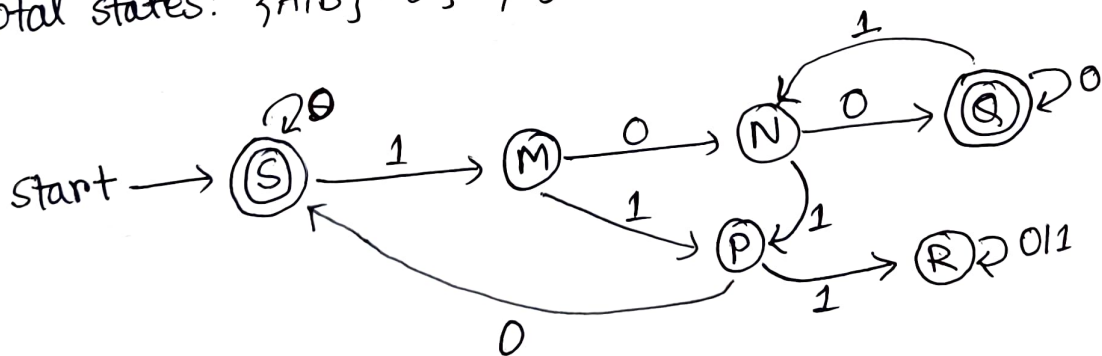
$\delta(P, 0) = \{A\} \xrightarrow{\epsilon} \{A, B\} = S$

$\delta(P, 1) = \emptyset = R$

$\delta(Q, 0) = \{A, B, C\} \xrightarrow{\epsilon} \{A, B, C\} = Q$

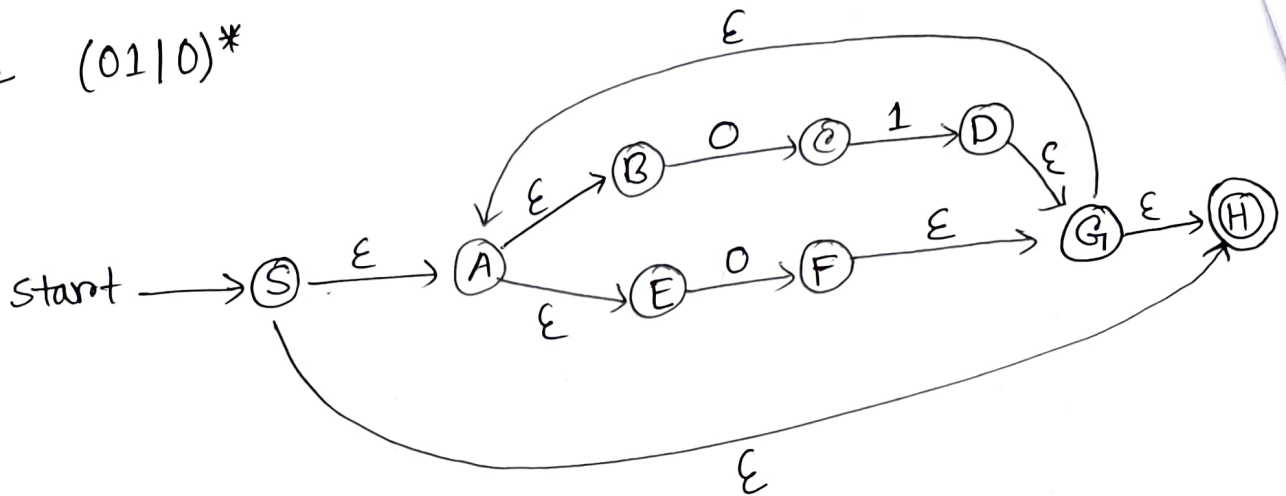
$\delta(Q, 1) = \{C, B\} = \{B, C\} = N$

Total states: $\{A, B\} = S$; $\{C\} = M$; $\{B, C\} = N$; $\{B\} = P$; $\{A, B, C\} = Q$



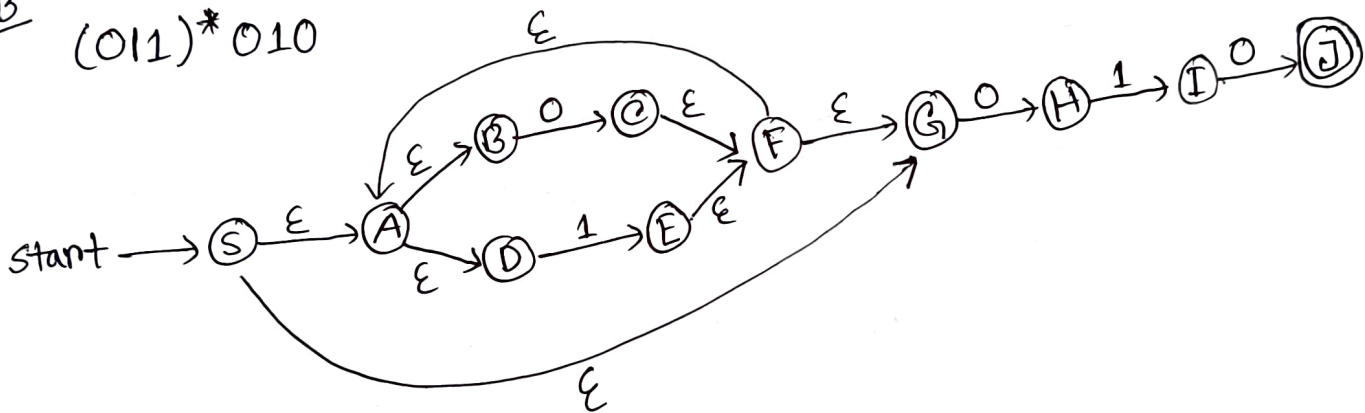
⑤

A $(01|0)^*$

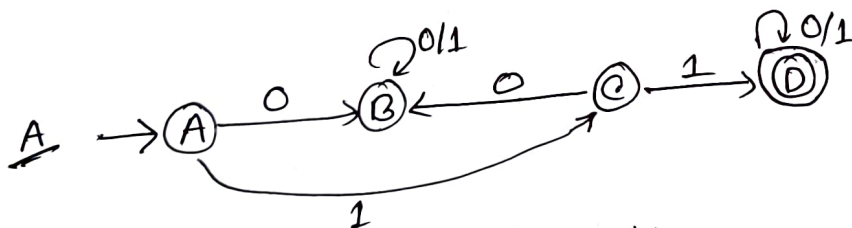


B

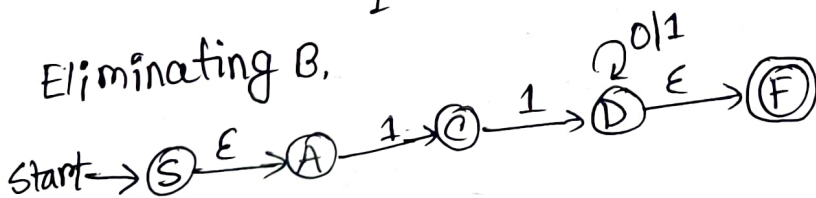
$(011)^* 010$



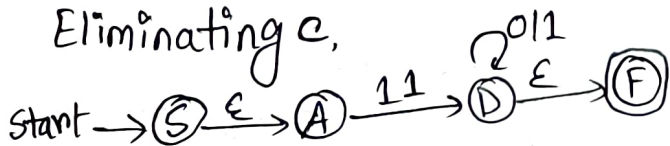
⑥



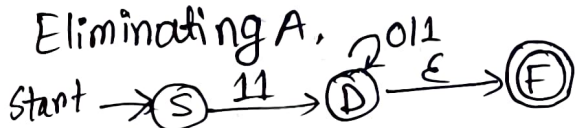
Eliminating B,



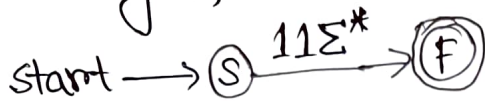
Eliminating C,



Eliminating A,

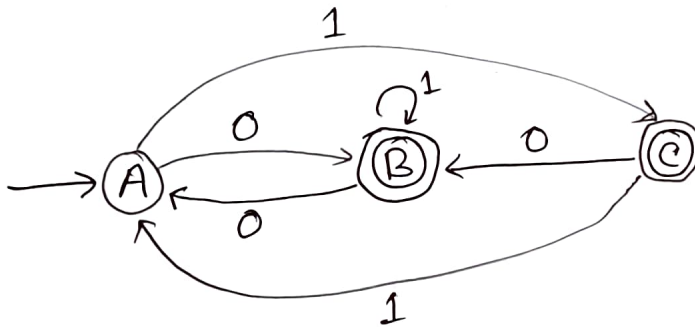


Eliminating D,

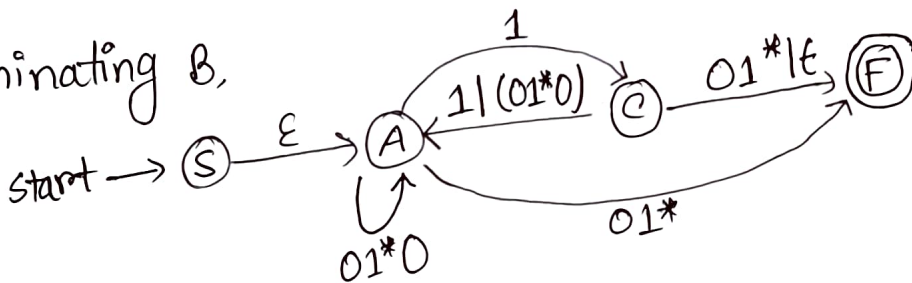


$$RE = 11\Sigma^*$$

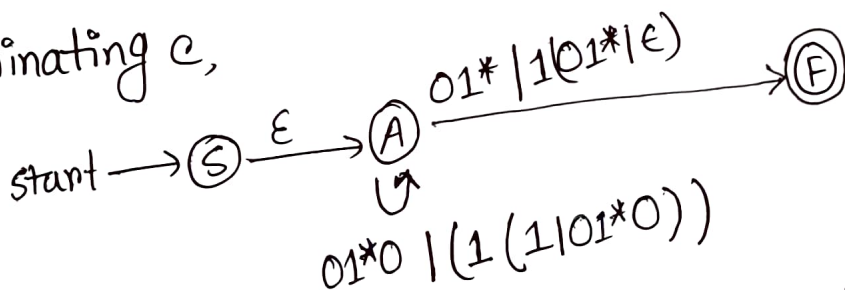
(B)



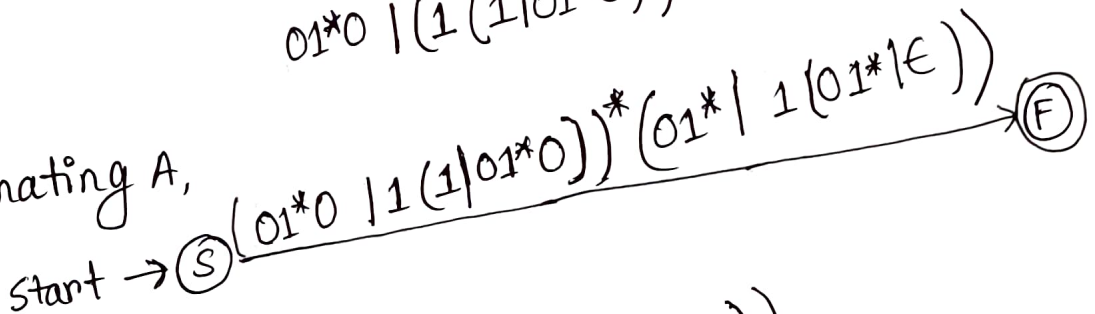
Eliminating B,



Eliminating C,



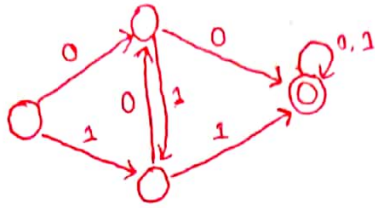
Eliminating A,



$$RE = (01^*0 | 1(1|01^*0))^* (01^* | 1(01^*|\epsilon))$$

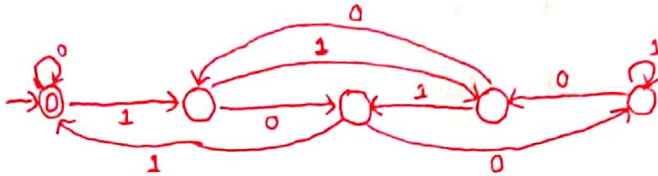
1/

a)

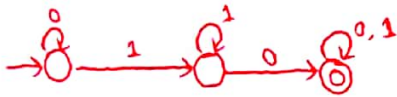


b) see Example 1.13 from the book.

c)



d)



2/

a) $(001)^* (01001\epsilon)$

b) $0\Sigma^*0 \mid 1\Sigma^*1 \mid 0 \mid 1 \mid \epsilon$

c) see 2(b) of the assignment.

d) $0^* (1^* 000^*)^* 1^* 0^*$

3/

a)

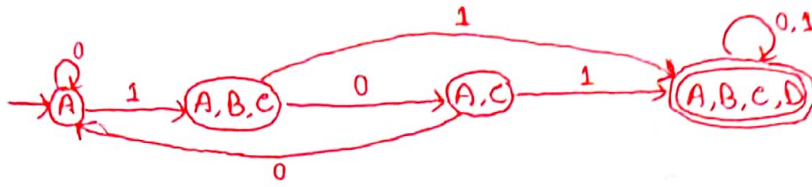


b) see Example 1.30 from the book.

c) see Example 1.33 from the book.

4/

a)



b) See Example 1.41 from the book.

5/

a) See Example 1.56 from the book.

b) See Example 1.58 from the book.

6/

a) $11\varepsilon^*$

b) See Example 1.68 from the book.

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CSE331 : Automata and Computability

Assignment 02

Summer 2024

Q1. Which of the following statements about regular expressions is true? (2 points)

1. Regular expressions can describe only Regular Languages.
2. Regular expressions can describe only Context Free Languages.
3. Regular expressions can describe both Regular Language and Context Free Languages.

Write an explanation for your answer.

Answer : Regular languages are recognized by finite automata, which have limited amount of memory. However, Context Free Languages require unbounded amount of memory, so regular expressions can't describe Context Free Languages.

Q2. Which is the correct regular expression for the language $L = \{w \in \{0,1\}^*: 0^n, \text{ where } n \text{ is even}\}$? (2 points)

1. $0(00)^*$
2. $(00)^*$
3. $0^*(00)^*0^*$
4. None of the above

Write an explanation for your answer.

Answer : Since n is even, the number of 0s in the language will also be even and $(00)^*$ contains even number of 0s.

Q3. Which is the correct regular expression for the language $L = \{w \in \{0,1\}^*: 0^n1^n, \text{ where } n \text{ is even}\}$? (2 points)

1. $(00)^*(11)^*$

2. $((0 \cup 1)(0 \cup 1))^*$
3. $(00 \cup 11)^*$
4. None of the above

Write an explanation for your answer.

Answer : Since, every option has $*$ above it, none can guarantee that in the string the number of 0s and 1s will be same.

Q4. You are given two languages, A and B. You can construct a regular expression for A. However, it is impossible to construct a regular expression for B. Now read the options below. (2 points)

- a) A is a regular language
- b) B is a regular language.
- c) B is a nonregular language.
- d) $A \cup B$ is a regular language. Choose the best answer from below.

1. a and b
2. a and c
3. a and d
4. a, b and d
5. a, c and d

Write an explanation for your answer.

Answer : Regular expressions describe regular language. Since I can construct regular expression for A but not for B, A is a regular language, and B is a nonregular language. $A \cup B$ will be regular if and only if A and B are regular. Though, $A \cup B$ they are closed under union, but when one of the languages is nonregular, the union will be nonregular.

Q5. What does the regular expression $(0 \cup 1)^* 0 (0 \cup 1)^*$ describe? (2 points)

1. $L = \{w \in \{0,1\}^*: w \text{ contains exactly one } 0\}$
2. $L = \{w \in \{0,1\}^*: w \text{ contains at least three } 0\}$
3. $L = \{w \in \{0,1\}^*: w \text{ contains at least one } 0\}$
4. $L = \{w \in \{0,1\}^*: w \text{ contains at most one } 0\}$ Write an explanation for your

answer.

Answer : In the given regular expression $(0 \cup 1)^* 0 (0 \cup 1)^*$, the expression $(0 \cup 1)^*$ can generate any number of 0s or 1s or empty. However, there is a 0 in the middle that always exists, which means that w contains at least one 0.

Q6. $(0 \cup 1)^* (0 (0 \cup 1) 0 \cup 1 (100 \cup 001 \cup 11^*)) 00$

Write down the shortest nonempty string generated by the given expression. Write down the string only. (2 points)

Write an explanation for your answer.

Answer : As we are searching for shortest nonempty strings, let $(0 \cup 1)^*$ be empty. Then $0 (0 \cup 1) 0$ becomes either 000 or 010. Next, $1 (100 \cup 001 \cup 11^*)$ simplifies to 11. So, $(0 \cup 1)^* (0 (0 \cup 1) 0 \cup 1 (100 \cup 001 \cup 11^*)) 00$ simplifies to $((000 \cup 11) \cup (010 \cup 11))00 = 1100$. Therefore, the shortest nonempty string is 1100.

Q7. $(00 \cup 01) (101)^* ((00)^* \cup 110)$

What is the shortest string generated by the regular expression with exactly three 0s? Write down the string. (2 points)

Write an explanation for your answer.

Answer : As we are searching for shortest string with exactly three 0s, from $(00 \cup 01)$, we choose 01, it gives one 0. Let $(101)^*$ be empty. Next, from $((00)^* \cup 110)$, we choose 00 from $(00)^*$, which provides two 0s, making a total of three 0s. Therefore, the shortest string with exactly three 0s is 0100.

Q8. Does the string 00100001011 belong to the regular expression $0 (0 \cup 1)^* 01 (0 \cup 1)^* ((0 \cup 1)^*)^* 11 (0 \cup 1)^* 1$? (2 points)

1. Yes
2. No

Write an explanation for your answer.

Answer : If we look at the regular expression, from the part $11 (0 \cup 1)^* 1$, even if we consider $(0 \cup 1)^*$ as empty, there will still be three 1s. However, in the string 00100001011, we see only two 1s at the end, so it doesn't fit the regular expression.

Q9. Let's, $\Sigma = \{a, b\}$. You are given a regular expression, $a(\Sigma^* a^+ \epsilon) \cup b(\Sigma^* b^+ \epsilon)$, what is the correct language for this regex? Here, $a^+ = aa^*$.

Note, for a regular language L, the regular expression will be correct if and only if it generates all the strings, $w \in L$, and doesn't generate any string, $w \notin L$. (2 points) a) w starts and ends with the same symbols

b) w contains either at least two a or at least two b

c) w contains substring ab

d) w contains an a followed by a b

e) w contains equal numbers of ab and ba Choose the best answer from below.

1. a
2. a and b
3. c and d
4. a and e
5. a, b and e

Write an explanation for your answer.

Answer : If we look at the regular expression $a(\Sigma^* a^+ \epsilon) \cup b(\Sigma^* b^+ \epsilon)$, we can see that if a string starts with a, it ends with a, and if it starts with b, it ends with b. Since, the string starts with a or b and ends with a^+ or b^+ , it contains either at least two a or at least two b. Then, Σ^* may not appear at all, and even if it does, it does not imply that w must contain the substring ab, an a followed by a b or equal numbers of ab and ba.

Q10. Which of the following Regular expressions are equivalent to $a(a^* b^* \cup b^+)b$ (2 points) a)

a) $a^*(ab^+)$

b) $a^+(ab^+)$

c) a^+b^+

d) $a(aa^*b^*)b$

e) $a^*(a^*b^+ \cup b^*)b$

Choose the best answer from below.

1. a and b

2. c and d
3. d and e
4. b and c

Write an explanation for your answer.

Answer : b, c are correct because they ensure strings must start with a and end with b.

Q11. $(01 \cup 1010 \cup 1)^* (00 \cup 010)$

The length of a string, w can be expressed by $|w|$. Find out how many distinct strings can be generated using the given regular expression such that $3 \leq |w| \leq 5$. Write a numeric value only.

(2 points)

Write an explanation for your answer.

Answer : Possible distinct strings of length between 3 and 5 are : 010, 1+00 = 100, 01+00 = 0100, 1+010 = 1010, 11+00 = 1100, 01+010 = 01010, 11+010 = 11010, 1+01+00 = 10100. The total number of distinct strings is 8.

Q12. You have a regular expression $(0 \cup 1)(0 \cup 1)00^*(0 \cup 1)^*1^*$. And another regular expression $(00 \cup 11 \cup 10 \cup 01)01^*0^*(1 \cup 0 \cup \epsilon)^*0^*$. Are these two the same? (2 points)

1. Yes
2. No

Write an explanation for your answer.

Answer : In the part '00*' of the first regular expression, it indicates that zero or more 0s can occur. In the part '01*0*' of the second regular expression, it indicates that zero or more 0s can occur, but any number of 1s can appear between them, which is not possible in the first regular expression. Additionally, the first regular expression can end with any number of 1s, while the second regular expression can end with any number of 0s. Essentially, they generate different sets of strings.

Q13. Let's $\Sigma = \{0,1\}$. What is the regular expression for the strings that have even length and start with 1? (2 points) a) $1(0 \cup 1)^*$

b) $(10 \cup 11)((0 \cup 1)(0 \cup 1))^*$

c) $(1(0 \cup 1))^*$

d) $1(0 \cup 1)((0 \cup 1)(0 \cup 1))^*$

e) $1(0 \cup 1)(00 \cup 01 \cup 11 \cup 10)^* \cup \epsilon$ Select the best answer.

1. d and e
2. b and e
3. **b and d**
4. b, d and e
5. a, c and e

Write an explanation for your answer.

Answer : Option b starts with 10 or 11, ensuring that the string starts with 1 and has an even length. It then continues with an even number of symbols. Option d is similar to option b. However, the other options do not require the strings to start with 1 or maintain an even length.

Q14. $L = \{w \in \{0,1\}^*: w \text{ doesn't contain } 001\}$

Note, for a regular language L , the regular expression will be correct if and only if it generates all the strings, $w \in L$, and doesn't generate any string, $w \notin L$. Which of the following Regular Expressions generate L :

- $(1^*(01)^*)^*(\epsilon \cup 0 \cup 000^*)$ (1 point)
 1. **Correct**
 2. Incorrect
- $(1^*(0 \cup \epsilon)1^*)^*$ (1 point)
 1. Correct
 2. **Incorrect**
- $(1 \cup 10)^*(\epsilon \cup 0 \cup 000^*)$ (1 point)
 1. Correct
 2. **Incorrect**
- $(1 \cup 01)^*0^*$ (1 point)
 1. **Correct**
 2. Incorrect
- $(1^*(10)^*)^*0^*$ (1 point)
 1. **Correct**
 2. Incorrect

Q15. $(0 \cup \varepsilon)(10)^*(1 \cup \varepsilon)$

Note, for a regular language L , the regular expression will be correct if and only if it generates all the strings, $w \in L$, and doesn't generate any string, $w \notin L$.

Which of the following regular languages are generated by the given regular expression

- $L = \{ w \in \{0,1\}^* : w \text{ doesn't contain } 00 \}$ (1 point)
 - 1. Correct
 - 2. Incorrect
- $L = \{ w \in \{0,1\}^* : 0 \text{ and } 1 \text{ alternates in } w \}$ (1 point)
 - 1. Correct
 - 2. Incorrect
- $L = \{ w \in \{0,1\}^* : w \text{ doesn't contain either } 00 \text{ or } 11 \}$ (1 point)
 - 1. Correct
 - 2. Incorrect
- $L = \{ w \in \{0,1\}^* : \text{The period of } w \text{ is either } 01 \text{ or } 10 \}$
The period of a string is the smallest prefix, when repeated one or more times, can recreate the entire string. (1 point)
 - 1. Correct
 - 2. Incorrect

All answers are not correct.

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CSE331 : Automata and Computability

Assignment 03

Summer 2024

Q1. Which of the following statements about Context Free Grammar (CFG) is true? (2 points)

1. Context Free Grammar (CFG) can describe only Context Free Languages.
2. Context Free Grammar (CFG) can describe all the Non Regular Languages and Regular Languages.
3. Context Free Grammar (CFG) can describe both Context Free Languages and Regular Language.

Write an explanation for your answer.

Answer : CFG can describe both context-free languages and regular languages, as regular languages are a subset of context-free languages.

Q2. Consider the following languages.

A) $L = \{ w \in \{a,b,c\}^* : a^n b^n c^n, \text{ where } n \geq 0 \}$

B) $L = \{ w \in \{0,1\}^* : ww \}$

C) $L = \{ w \in \{0,1\}^* : ww^R \}$ [w^R means reverse of w . For example, if $w = 10110$, then $w^R = 01101$.]

D) $L = \{ w \in \{a,b,c\}^* : a^i b^j c^k, \text{ where } 0 \leq i \leq j \leq k \}$

Which languages are Context Free Languages?

1. Only C
2. B and C
3. C and D
4. A, B and C

Write an explanation for your answer.

Answer: Option C is a palindromes problem.

Q3. Let, L is a context-free language. Then L^* will also be a context-free language.

The statement is

1. True
2. False
3. Not enough information to answer

Write an explanation for your answer.

Answer: Kleene star of a context-free language is always context-free.

Q4. Consider the following Context Free Grammar.

$S \rightarrow AB \mid BC$

$A \rightarrow aA \mid Aa \mid a \mid \epsilon$

$B \rightarrow bBb \mid bb$

$C \rightarrow c \mid \epsilon$

Choose the correct regular expression for the language covered by the given CFG

1. $a^*(bb)^* \mid (bb)^*c^*$
2. $a^+(bb)^* \mid (bb)^+c^?$
3. $a^+(bb)^+ \mid (bb)^+(c|\epsilon)$
4. $a^*(bb)^+ \mid (bb)^+(c|\epsilon)$
5. $a^+(bb)^* \mid (bb)^*(c|\epsilon)$

Write an explanation for your answer.

Answer: A produces a^* , B produces sequences of bb and C can either produce c or ϵ .

Q5. $A \rightarrow 1A \mid 0A1 \mid 01$

Which of the following string can be generated by the given CFG.

There could be more than one correct answer, choose all of them.

1. 000111
2. 001111
3. 001011
4. 0100111
5. 1000111

Write an explanation for your answer.

1. $A \rightarrow 0A1 \rightarrow 00A11 \rightarrow 000111$
2. $A \rightarrow 0A1 \rightarrow 01A1 \rightarrow 010A11 \rightarrow 0100111$
3. $A \rightarrow 1A \rightarrow 10A1 \rightarrow 100A11 \rightarrow 1000111$

Q6. You are given a Context Free Grammar.

$$S \rightarrow AB$$
$$A \rightarrow aA \mid \varepsilon$$
$$B \rightarrow Bb \mid \varepsilon$$

Now consider the following languages.

- a) $L = \{ w \in \{a,b\}^* : a^n b^m, \text{ where } n, m \geq 0 \}$
- b) $L = \{ w \in \{a,b\}^* : a^n b^n, \text{ where } n \geq 0 \}$
- c) $L = \{ w \in \{a,b\}^* : \text{number of } a \text{ and number of } b \text{ is unequal in } w \}$
- d) $L = \{ w \in \{a,b\}^* : \text{all } a \text{ in } w \text{ always precede } b \}$

Which of the following statements are true regarding the language generated by this grammar?

Note, for a language L , the CFG will be correct if and only if it can parse all the strings, $w \in L$, and doesn't parse any string, $w \notin L$.

- 1. Only A
- 2. Only B
- 3. A and C
- 4. A and D
- 5. B and D

Write an explanation for your answer.

Answer : As $S \rightarrow AB$ it can generate $a^n b^m$ successfully and all a always precede b .

Q7. $L = \{ w \in \{a,b,c\}^* : a^m b^n c^k \mid n, m \geq 0 \ \& \ k=3m+2 \}$

Let's say, we have the following four CFGs labeled as (A) to (D).

CFG A:

$$S \rightarrow aScc \mid Y$$
$$Y \rightarrow bY \mid ccc$$

CFG B:

$$S \rightarrow aSccc \mid Y$$
$$Y \rightarrow bY \mid cc$$

CFG C:

$S \rightarrow Xcc$

$X \rightarrow aXccc \mid Z$

$Z \rightarrow bZ \mid \varepsilon$

CFG D:

$S \rightarrow aSccc \mid bS \mid cc$

What will be the correct CFG for the language L?

Note, for a language L, the CFG will be correct if and only if it can parse all the strings, $w \in L$, and doesn't parse any string, $w \notin L$.

1. Only B
2. Only C
3. A and C
4. A and D
5. B and C
6. B and D

Write an explanation for your answer.

Answer : B and C generates all the strings where number of c's = 3(number of a's) + 2 and they can parse all the strings, $w \in L$, and doesn't parse any string, $w \notin L$.

Q8. $L = \{ w \in \{a,b\}^* : w = a^i b^j \text{ where } i = 2+j, j \geq 0 \}$

Let's say, we have the following four CFGs labeled as (A) to (C).

CFG A:

$S \rightarrow aaN$

$N \rightarrow aNb \mid \varepsilon$

CFG B:

$S \rightarrow aSb \mid aa$

CFG C:

$S \rightarrow aSb \mid N$

$N \rightarrow aaN \mid \varepsilon$

What will be the correct CFG for the language L?

Note, for a language L , the CFG will be correct if and only if it can parse all the strings, $w \in L$, and doesn't parse any string, $w \notin L$.

1. A and C
2. A and B
3. B and C
4. A, B and C

Write an explanation for your answer.

Answer : CFG A & B always generates those strings where $i = 2 + j$ and they can parse all the strings, $w \in L$, and doesn't parse any string, $w \notin L$.

Q9. Consider the following two Context Free Grammars. Do these two grammars (G1 and G2) represent the same language?

G1:

$S \rightarrow AB$

$A \rightarrow 1A00 \mid C$

$C \rightarrow 0C \mid 0$

$B \rightarrow 1B \mid \epsilon$

G2:

$S \rightarrow ABC$

$A \rightarrow 1A00 \mid \epsilon$

$B \rightarrow 0B \mid 0$

$C \rightarrow 1C \mid \epsilon$

1. Yes
2. No

Write an explanation for your answer.

Answer : G1 and G2 always generates same strings, so they represent the same language.

Q10. $L = \{ w \in \{a,b,c\}^* : a^m b^n c^k \mid n = m/2 \text{ or } n = k - 2 ; \text{ where } n \geq 0, m \text{ is even \& } k \geq 2 \}$

Let's say, we have the following 4 CFGs labeled as (A) to (D).

CFG A:

$S \rightarrow XY \mid PQ$

$X \rightarrow aXbb \mid \epsilon$

$Y \rightarrow ccY \mid c$

$$P \rightarrow aP \mid b$$

$$Q \rightarrow bQc \mid c$$

CFG B:

$$S \rightarrow XY \mid PQ$$

$$X \rightarrow aXb \mid \varepsilon$$

$$Y \rightarrow cYb \mid c$$

$$P \rightarrow aP \mid \varepsilon$$

$$Q \rightarrow bQcc \mid c$$

CFG C:

$$S \rightarrow XY \mid PQ$$

$$X \rightarrow aXb \mid c$$

$$Y \rightarrow cYb \mid \varepsilon$$

$$P \rightarrow aP \mid \varepsilon$$

$$Q \rightarrow bQcc \mid c$$

CFG D:

$$S \rightarrow XY \mid PQ$$

$$X \rightarrow aaXb \mid \varepsilon$$

$$Y \rightarrow cY \mid cc$$

$$P \rightarrow aaP \mid \varepsilon$$

$$Q \rightarrow bQc \mid cc$$

What will be the correct CFG for the language L?

Note, for a language L, the CFG will be correct if and only if it can parse all the strings, $w \in L$, and doesn't parse any string, $w \notin L$.

1. A
2. B
3. C
4. D

Write an explanation for your answer.

Answer : D always generates those strings where $n = m/2$ or $n = k - 2$; where $n \geq 0$, m is even & $k \geq 2$.

Q11. $L = \{w \in \{0,1\}^* : w_1w_2w_3 : \text{where } |w_1| = |w_2| \text{ or } |w_3| = |w_1|\}$

Let's say, we have the following 4 CFGs labeled as (A) to (D).

CFG A:

$S \rightarrow RQ \mid P$

$P \rightarrow 0P1 \mid 1P0 \mid \varepsilon$

$Q \rightarrow 0Q \mid 1Q \mid \varepsilon$

$R \rightarrow 0R0 \mid 1R1 \mid 0R1 \mid 1R0 \mid Q$

CFG B:

$S \rightarrow PR \mid Q$

$P \rightarrow 0P0 \mid 1P1 \mid \varepsilon$

$Q \rightarrow 0Q \mid 1Q \mid \varepsilon$

$R \rightarrow 0R0 \mid 1R1 \mid 0R1 \mid 1R0 \mid Q$

CFG C:

$S \rightarrow PQ \mid R$

$P \rightarrow 0P0 \mid 1P1 \mid 0P1 \mid 1P0 \mid \varepsilon$

$Q \rightarrow 0Q \mid 1Q \mid \varepsilon$

$R \rightarrow 0R0 \mid 1R1 \mid 0R1 \mid 1R0 \mid Q$

CFG D:

$S \rightarrow PQ \mid R$

$P \rightarrow 0P0 \mid 1P1 \mid 0P1 \mid 1P0 \mid \varepsilon$

$Q \rightarrow 0Q \mid 1Q$

$R \rightarrow 0R0 \mid 1R1 \mid 0R1 \mid 1R0 \mid P$

What will be the correct CFG for the language L?

Note, for a language L, the CFG will be correct if and only if it can parse all the strings, $w \in L$, and doesn't parse any string, $w \notin L$.

1. A and C
2. B and C
3. C and D
4. A, B and C
5. A, B, C and D

Write an explanation for your answer.

Answer : A, B and C always generates those strings where $|w_1| = |w_2|$ or $|w_3| = |w_1|$

Q12. $L = \{w \in \{0,1\}^* : w_1\#w_2\#w_3 : \text{where } w_3 = w_1^R\}$

Given a string w over some alphabet Σ , let w^R be its reverse. For example, if $w = 10110$, then $w^R = 01101$.

Let's say, we have the following 4 CFGs labeled as (A) to (C).

CFG A:

$$S \rightarrow 1S0 \mid 0S1 \mid \#P\#$$
$$P \rightarrow 0P \mid 1P \mid \varepsilon$$

CFG B:

$$S \rightarrow 0S \mid 1S \mid \#P\#$$
$$P \rightarrow 0P0 \mid 1P1 \mid \varepsilon$$

CFG C:

$$S \rightarrow 0S0 \mid 1S1 \mid \#P\#$$
$$P \rightarrow 0P \mid 1P \mid \varepsilon$$

What will be the correct CFG for the language L?

Note, for a language L, the CFG will be correct if and only if it can parse all the strings, $w \in L$, and doesn't parse any string, $w \notin L$.

1. A
2. B
3. C

Write an explanation for your answer.

Answer : C always generates those strings ($w_1\#w_2\#w_3$) where $w_3 = w_1^R$

Q13. $L = \{ w \in \{0,1\}^* : w \in \Sigma^* \mid \text{every prefix of } w \text{ has at least as many 0's as 1's} \}$

Clarification: L states "Every prefix of w contains equal or more 0s than 1s."

Select the correct CFG for L:

A) $S \rightarrow 0S1S \mid 0S \mid \varepsilon$

B) $S \rightarrow SS \mid 0S1 \mid 1 \mid \varepsilon$

C) $S \rightarrow SS \mid 0S1 \mid 0 \mid \varepsilon$

D) $S \rightarrow 0S1S \mid 1S \mid \varepsilon$

A prefix is a string consisting of several first letters of a given string, without any reorders. An empty prefix is also a valid prefix. For example, the string "abcd" has 5 prefixes: empty string, "a", "ab", "abc" and "abcd".

1. a and d
2. a and b
3. a and c
4. a, b and d
5. a, b and c
6. a, b, c and d

Write an explanation for your answer.

Answer : a and c always generate those strings where every prefix of w has at least as many 0's as 1's.

Q14. Let $\Sigma = \{a, b\}$ and let $L = \{ a^n b^m \mid n, m \in \mathbb{N} \text{ and } n \leq m \leq 5n \}$. The CFG for L is given below.

Here is a CFG of the language

$S \rightarrow aSb? \mid \varepsilon$

$A \rightarrow b \mid \varepsilon$

What will be the '?' such that the grammar can describe the language, L correctly.

Fill out the question mark with one or more variables/terminals. Write down the missing string only.

Write an explanation for your answer.

Answer: Replace the ? with AAAA and it will correctly describe the language L

Q15.

$E \rightarrow E + P \mid P$

$P \rightarrow P * Q \mid Q$

$Q \rightarrow id$

Is the grammar ambiguous?

1. Yes
2. No

Write an explanation for your answer.

Answer : This grammar has only 1 Left Most Derivation (LMD), 1 Right Most Derivation (RMD) and 1 Parse Tree (PT).

Q16. You are given two Context Free Grammar G1 and G2.

G1:

$S \rightarrow (S)S \mid \varepsilon$

G2:

$S \rightarrow (S) \mid SS \mid \varepsilon$

Now, consider the string $w = ((()((()))((()((()))))())$

Which of the following statements is true?

1. w can be generated using only G1
2. w can be generated using only G2
3. w can be generated using G1 and G2
4. w cannot be generated using either G1 or G2

Write an explanation for your answer.

Answer : G1 and G2 both can generate the string w because both can handle balanced parentheses.

Q17. Consider the two Context free grammars G1 and G2. Which of the following is $L(G1) \cap L(G2)$?

G1:	G2:
$S \rightarrow ASA$	$S \rightarrow ASA$
$S \rightarrow 0$	$S \rightarrow \varepsilon$
$A \rightarrow 0 \mid 1$	$A \rightarrow 0 \mid 1$

1. $L = \{w \in \Sigma^* \mid w \text{ contains } 0 \text{ in the middle.}\}$
2. $L = \{w \in \Sigma^* \mid \text{length of } w \text{ is even.}\}$
3. $L = \{w \in \Sigma^* \mid w \text{ is a even length string with } 0 \text{ in the middle.}\}$
4. $L = \{w \in \Sigma^* \mid w \text{ is a odd length string with } 0 \text{ in the middle.}\}$
5. None of the above.

Write an explanation for your answer.

Answer : Length of G1 is odd and length of G2 is even so their intersection is not possible.

Q18. Consider the following Context Free Grammar, G.

G:

$P \rightarrow XP1 \mid XQ0$

$Q \rightarrow XQ1 \mid XR0$

$R \rightarrow XRX \mid \#$

$X \rightarrow 0 \mid 1$

Now, answer the following questions.

What is the shortest string can be parsed from the grammar, G? If there are multiple correct answer write only one of those.

Answer : 01#00

Write an explanation for your answer.

Answer : $P \rightarrow XQ0 \rightarrow 0Q0 \rightarrow 0XR00 \rightarrow 01R00 \rightarrow 01\#00$

The length of a string, w can be expressed by $|w|$.

Find out how many distinct strings can be generated using the given Context Free Grammar, G, such that $9 \leq |w| \leq 11$. Write the numeric value only.

Answer : 1008

Write an explanation for your answer.

Answer : Using the grammar production rules, by counting and summing all the possible combinations, we obtain 1008 distinct strings with lengths between 9 and 11.

Q19. Consider the following Context Free Grammar, G.

G:

$S \rightarrow aaS \mid abS \mid baS \mid bbS \mid X$

$X \rightarrow aaY \mid baY$

$Y \rightarrow aY \mid bY \mid \epsilon$

Write a eight length string that starts with "bb" and has only one parse tree. Write the string only.

Answer : bbababba

Write an explanation for your answer.

Answer : $S \rightarrow bbS \rightarrow bbabS \rightarrow bbababS \rightarrow bbababX \rightarrow bbababbaY \rightarrow bbababba$

Q20. $L = \{w \in \{a,b,c\}^* : a^i b^j c^k \mid i, j, k \geq 0 \text{ and if } i > j \text{ then } k = i - j, \text{ else } k = 0\}$

Which of the following CFGs generates L?

CFG A:

$S \rightarrow AB \mid AC$

$A \rightarrow aA \mid \varepsilon$

$B \rightarrow bB \mid b$

$C \rightarrow aCc \mid \varepsilon$

CFG B:

$S \rightarrow AB \mid A$

$A \rightarrow aA \mid a \mid \varepsilon$

$B \rightarrow bBc \mid b$

CFG C:

$S \rightarrow A \mid B$

$A \rightarrow aAb \mid Ab \mid \varepsilon$

$B \rightarrow aBc \mid aCc$

$C \rightarrow aCb \mid \varepsilon$

CFG D:

$S \rightarrow AB \mid AC$

$A \rightarrow aAb \mid \varepsilon$

$B \rightarrow bB \mid \varepsilon$

$C \rightarrow aCc \mid \varepsilon$

Note, for a language L , the CFG will be correct if and only if it can parse all the strings, $w \in L$, and doesn't parse any string, $w \notin L$.

1. A
2. B
3. C
4. D

Write an explanation for your answer.

Answer : C always generates those strings ($a^i b^j c^k$) where $i, j, k \geq 0$ and if $i > j$ then $k = i - j$, else $k = 0$

Q21. $L = \{w \in \{a, b\}^* : \text{each } b \text{ in } w \text{ is followed by at least two } a\}$

Note, for a language L , the CFG will be correct if and only if it can parse all the strings, $w \in L$, and doesn't parse any string, $w \notin L$.

Which of the following Context Free Grammar can generate L ?

$S \rightarrow XY$

$X \rightarrow aX \mid \varepsilon$

$Y \rightarrow ZY \mid \varepsilon$

$Z \rightarrow PQ$

$P \rightarrow ba$

$Q \rightarrow aQ \mid a$

1. Correct
2. Incorrect

$P \rightarrow aP \mid bQ \mid \varepsilon$

$Q \rightarrow aR \mid bT$

$R \rightarrow aS \mid bT$

$S \rightarrow aS \mid bQ \mid \varepsilon$

$T \rightarrow aT \mid bT$

1. Correct
2. Incorrect

$S \rightarrow AbR \mid A$

$A \rightarrow aA \mid \varepsilon$

$R \rightarrow aAabR \mid A$

1. Correct
2. Incorrect

ALL ANSWERS ARE NOT CORRECT